COVID-19 Vaccination & Mortality Analysis

Exploratory Data Analysis and Visualization (MSBA 625 - 001)

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Executive Summary

The United States experienced a severe COVID-19 surge in January 2022, recording 943,113 national daily cases and 3,619 daily deaths at the height of the Omicron wave. Using the CDC longitudinal data from December 2020 through January 2022, this analysis tracks state by state case growth, mortality, vaccination rates, and booster coverage to understand how the pandemic unfolded across the country. The US case Heat Map confirms that no state was spared, while California and Michigan led all states in new daily cases at 109K and 98K respectively. Despite the universal spread of infection, the Stability Scatter Plot shows that states exceeding 70% vaccination threshold consistently maintained lower daily death counts, proving that vaccines effectively reduced mortality even when they could not stop infection. The Dual-Encoding Vulnerability Map further exposes gaps between vaccine distribution and actual administration, showing that supply reaching a state did not always translate into supply reaching people. Most critically, the Risk Priority Matrix identifies states like New Jersey and Rhode Island as highly vulnerable, with high daily case growth but booster coverage well below the 50% priority threshold. These findings make clear that booster campaigns must be treated with the same urgency as initial rollouts, that distribution bottlenecks must be addressed through targeted outreach, and that future preparedness efforts should monitor case rates and vaccination coverage together in real time so that at-risk communities can be identified and supported before mortality rises.

Context & Motivation

The COVID-19 pandemic created unprecedented public health, educational, and resource allocation challenges across the United States. While the virus spread nationwide, its impact was not uniform because infection rates, mortality, and vaccination progress varied significantly by state and region. To address these disparities, this analysis utilizes a longitudinal dataset from the Centers for Disease Control and Prevention (CDC) covering December 2020 through January 2022. This data captures granular metrics including total doses distributed, the number of individuals fully vaccinated, and the daily growth of cases and deaths.

The primary motivation for this study is to move beyond national averages to uncover the state-level disparities that characterized the national rollout. By analyzing vaccination coverage in tandem with daily mortality and case surges, specifically during the rapid rise of the Omicron variant in early 2022, we can identify which populations remained most vulnerable despite the availability of vaccines. The central problem this analysis investigates involves identifying significant geographic variations in vaccination rates across the U.S. and evaluating how efficiently states convert distributed supplies into administered doses.

Understanding these trends is critically important because vaccination remains the primary defense against large-scale outbreaks. Identifying operational bottlenecks where high distribution does not correlate with high administration is essential for predicting future healthcare strain and informing the timing of economic reopening measures. The intended audience for this report includes public health administrators and state-level policy analysts. For these stakeholders, the stakes are exceptionally high because a failure to achieve equitable and timely vaccination coverage directly impacts public safety and jeopardizes the long-term economic stability of their regions.

Main Findings

The National Daily Cases and National Daily Deaths dashboards establish the scale of the crisis, with 943,113 new cases and 3,619 deaths recorded in a single day at the peak of the Omicron wave, signaling that the health system was under extraordinary strain and that decision makers needed immediate clarity in where to direct resources. The US Case Heat Map confirms that no state was spared, and the Top 10 States by Case Surge shows California and Michigan bearing the heaviest burden at 109K and 98K new daily cases respectively, which is important because it allows public health administrators to pinpoint which states required the most urgent intervention. With 61 million total vaccinations administered nationally at this point, the Stability Scatter Plot provides the most critical insight of the analysis: states that surpassed the 70% vaccination threshold consistently recorded lower daily death counts even as cases surged, proving that vaccines effectively separated infection from mortality and that increasing vaccination coverage was a viable strategy for protecting lives even during a highly transmissible wave. The Dual-Encoding Vulnerability Map adds another layer by revealing that several states received substantial vaccine supply but failed to convert it into administered doses, which matters because it shows that the problem was not always a lack of resources but a failure of delivery and outreach. The Risk Priority Matrix ties these threads together, identifying states like New Jersey and Rhode Island as critically vulnerable due to high daily case growth combined with booster coverage below the 50% priority threshold, leaving populations exposed in ways that full vaccination alone could not fix. Collectively, these visualizations tell a consistent story: the pandemic's impact was never uniform, and booster coverage was the defining factor in determining how well a state was able to limit mortality during the Omicron surge.

Conclusion

The data makes clear that vaccination coverage, and booster uptake specifically, was the most consequential factor separating states with manageable mortality from those that remained critically vulnerable. While Omicron proved no state could fully prevent infection, the mortality divide between high and low vaccination states was consistent and significant.

Policymakers and public health administrators should take away three priorities. First, booster campaigns must be treated with the same urgency as initial rollouts, as partial coverage, seen in states like New Jersey, leaves populations meaningfully exposed. Second, the gap between vaccine distribution and administration points to a logistics and outreach failure that must be addressed; supply reaching a state is not the same as supply reaching people. Third, future preparedness plans should monitor case rates and vaccination coverage together in real time, so vulnerable regions can be identified and supported before mortality rises.

Most importantly, this analysis challenges any strategy that evaluates a national campaign on aggregate numbers alone. The disparities documented here demand targeted, region specific intervention. Decision makers should move away from treating the U.S. as a uniform population and recognize that equitable public health outcomes require sustained, localized investment in the communities most at risk.

References

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